

Bill of Materials

EECE 5155 – IoT-Based Acoustic Monitoring System
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 Budget: \$50 USD (excluding MCU costs)

MCU (Excluded from Budget)

Item	Unit Cost	Qty	Total	Source
ESP8266 NodeMCU v3	\$6.00	2	\$12.00	Amazon/AliExpress

MCU costs excluded per assignment specification. 2 units required for 2-node MVP (glass room + solid wall deployment).

Sensor Nodes (2 nodes)

Item	Unit Cost	Qty	Total	Source
MAX9814 microphone breakout (Adafruit #1713)	\$7.95	2	\$15.90	Adafruit
USB-A to Micro-B cable 6 ft	\$2.50	2	\$5.00	Amazon
5V 1A USB wall adapter	\$2.00	2	\$4.00	Generic/Anker
ABS enclosure Hammond 1551 series	\$4.50	2	\$9.00	Amazon/Mouser
3M Command strips (damage-free mounting)	\$4.50	1	\$4.50	Amazon
Subtotal			\$38.40	

Validation and Development Equipment

Item	Unit Cost	Qty	Total	Source
Breadboard + jumper wire kit	\$10.00	1	\$10.00	Amazon
Subtotal			\$10.00	

Sound Level Meter (calibration reference): The BAFX3608 Digital Sound Level Meter (\$30, ANSI S1.4-2014 Class 2, ± 1.5 dBA) specified in PES v1 Section 6 exceeds the remaining budget after sensor node components. Mitigation strategy: borrow from Northeastern ECE stockroom or EXP makerspace (request submitted Week 1). Fallback: NIOSH SLM app (CDC, free) on calibrated smartphone – accuracy ± 2 dBA on modern devices, sufficient for ± 3 dBA system target with documented uncertainty increase.

Budget Summary

Category	PES v1 (excl. MCU)	Revised (\$50 budget)
2× sensor nodes (excl. MCU)	\$50.00	\$38.40
Validation equipment	\$55.00	\$10.00
Spare components (contingency)	\$16.00	—
Total (excl. MCU)	\$121.00	\$48.40
Remaining budget		\$1.60

Total system cost including MCU: $\$48.40 + \$12.00 = \$60.40$, within the original PES v1 \$200 budget constraint (Section 4, Constraint #1) with \$139.60 remaining for contingency and sound meter if lab loan unavailable.

Component Justifications

MAX9814 microphone breakout – \$7.95 (Adafruit #1713)

Electret microphone with integrated AGC amplifier. SNR 61 dBA provides 12 dBA margin above 45 dBA detection threshold. Analog output compatible with ESP8266 `analogRead()` – 2-hour implementation vs 8+ hours for I2S digital alternative. Selected over cheaper alternatives per PES v1 Section 4 Conflict 2: implementation simplicity prioritised over marginal cost savings given 10-week timeline constraint.

Hammond 1551 ABS enclosure – \$4.50

Downgraded from Hammond 1591 series (\$8.00 in PES v1) to 1551 series (\$4.50) – smaller footprint appropriate for wall-mounted microphone sensor, saves \$7.00 for 2 nodes. Still professional appearance for institutional deployment (libraries, healthcare facilities). IP54 rating protects against dust and splashing adequate for indoor HVAC-controlled environment.

USB power supply – \$2.00 per node

Wall-powered via standard 5V/1A USB adapters (phone chargers repurposed). Per-node consumption: $89.1 \text{ mA} @ 5\text{V} = 445 \text{ mW}$ continuous (PES v1 Section 6). Annual electricity cost \$0.59/node at \$0.15/kWh – negligible. Eliminates battery constraints and enables 100% duty cycle.